SKYWAVE™
AN ORBCOMM COMPANY



ST 8100

Solar-powered, fully programmable terminal for satellite-cellular IoT applications.

The ST 8100 dual-mode satellite-cellular terminal supports cost-effective IoT solutions that track and monitor mobile and fixed assets in industries such as maritime, transportation, oil & gas, agriculture, water management and more. It's ideal for applications without external power, such as tracking small vessels or monitoring fixed assets in remote locations, thanks to its solar-powered battery.

Optimized asset tracking

With dual-mode connectivity, the ST 8100 enables least-cost routing—using cellular when available and satellite via SKYWAVE's OGx network when needed.

Easy to install

Simplify solution implementation with the ST 8100's small form factor, cableless installation and out-of-the-box support for SKYWAVE's suite of wireless sensors. Plus, it can power wired sensors with a 24V output and integrate with mobile apps for fishing reports, text messaging and email.

Customizable

The ST 8100 can run SKYWAVE's off-the-shelf terminal applications—or support custom applications that run directly on the device.

- **Solar powered**
- **Satellite and cellular connectivity**
- **Programmable**
- **Wireless or wired sensor support**
- **Rugged**

Specifications

Satellite communication

- Satellite service: OGx (two-way, global)
- Typical latency: <15 seconds, 100 bytes
- Elevation angle: +20° to +90°
- Frequencies:
  - *Rx 1525.0–1559.0 MHz;*
  - *Tx 1626.5–1660.5 MHzz*
- EIRP: <6.5 dBW

Cellular communication

- LTE: Cat M1/NB2 (B1, B2, B3, B4, B5, B8, B12, B13, B18, B19, B20, B25, B26, B28, B66, B85)
- 2G: 4-band

Dimensions

- 285 x 93 x 35 mm

Memory

- Lua code: PSRAM 8MB
- NVM: 16MB

Certifications (pending)

- CE, FCC/IC, FFA, PTCRB, Inmarsat type approval, IEC 60945, CID2, HERO
- RCM, ICASA, Anatel, IFT, Japan, FFA

Other interfaces

- Bluetooth v5.0 low energy module

Solar power and battery

- Solar panel 0.86W (nominal)
- Lithium-ion battery 1,100 mAh
- Discharge temperature range: -35°C to +75°C
- Charge temperature range:
  - *No connector: -20°C to +75°C*
  - *With connector: -35°C to +75°C*

External interfaces (Advanced variant only)

- 2 configurable inputs/outputs: Analog/digital/input/output
- 2 analog voltage inputs (0-15V)
- 2 analog current inputs (4-20mA)
- Serial: 1 RS-232; 1 RS-485/J1708; 1-Wire
- 2 Power outputs (24V, 30mA each)

GPS/Glonass/Beidou/Galileo

- Acquisition time:
  - *Hot: 1 second*
  - *Cold: 27–56 seconds*
- Accuracy: 1.5-4.0m CEP-horizontal
- Sensitivity:
  - *Acquisition: -148 dBm*
  - *Tracking: -167 dBm*
- Security: signal jamming detection

Accelerometer

- 3-axis accelerometer

Environmental

- Operating temperature: -35°C to +75°C
- Dust and water ingress:
  - *IP67/IPX9K*
- Vibration: SAE J1455 (Sec 4.10.4.2 fig 6–8); MIL-STD-810H (Sec 514.8)
- Shock: MIL-STD-810G (Sec 516.8)

Programming

- Lua scripting engine with core services. SDK with GUI development tools available. Lua software application and firmware upgradable over the air (SOTA, FOTA)
- Geofencing: 128 Polygons
- Data logger: 50,000 position reports
- Optional, configurable terminal apps

Order codes

- ST8100-1100-H —Advanced variant
- ST8100-0100-H — Standard variant
- ST101692-001 — ST 8100 Development Kit

Although we strive to ensure accuracy in all of our published specifications, actual field performance can vary depending on a variety of environmental, installation and usage factors, as well as third-party factors such as cellular providers. The specifications listed are approximations, and do not constitute binding statements or modify the terms and conditions of purchase or lease including, but not limited to, product operational limitations and warranties. All specifications are subject to change without notice. Please check www.skywave.com to ensure you have the latest version of these specifications.

Email: **sales@skywave.com** | Call: **1-800-672-2666**

Visit our website **www.skywave.com**

SKYWAVE, an ORBCOMM company, is a global provider of IoT solution enablement technology. We empower solution providers, system integrators and OEMs to serve their customers through satellite and cellular managed IoT networks and a complete application enablement platform. We provide a fully integrated ecosystem of purpose-built devices, data automation and connectivity services for high-reliability, low-data solutions. SKYWAVE is where IoT powers mission-critical applications for the transportation, agriculture, oil and gas and maritime industries.

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